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Supplementary Information: scientific judgments in autonomous research

Jiangjie Qiu[1,*], Yijun Li[1,*], Yaotian Yang[1,*], Honghao Chen[1], Wentao Li[1], Xiaonan Wang[1,†]

[1]Beijing Key Laboratory of Artificial Intelligence for Advanced Chemical Engineering Materials, State Key Laboratory of Chemical Engineering and Low-Carbon Technology, Department of Chemical Engineering, Tsinghua University, Beijing 100084, China.

* Jiangjie Qiu, Yijun Li and Yaotian Yang contributed equally.

† Correspondence: wangxiaonan@tsinghua.edu.cn.

Retained numerical evidence. Research cohorts and all failure dispositions remain separate.

Supplementary Information

Study organization

The broad study (300 campaigns), earlier selected notebook studies (24 sessions), and repeated process cohort (90 sessions) retain separate denominators and execution provenance. Reference batches, recommendation retests and fixed-recipe controls are not researcher sessions. The process cohort retains 84 complete chains, 1,079/1,080 source batches, 298/315 assessment stages and 86 valid forecast stages. Notebook reading covers all 284 revisions, but qualitative interpretation is by one analyst and selected cases do not establish mechanism prevalence.

Complete equilibrium process outcomes

Each row is one independent model session within a fixed world and information condition. O/A/M denote Opaque/Aligned/MisIndexed. A dash is a missing forecast. All M sessions belong to the extension cohort; O/A sessions belong to the first cohort. MAEs average the three equilibrium response channels. The other-nine group is not an in-distribution control.

World	Arm	Repeat	Batches	Other-nine MAE	Dilute MAE	Dilute coverage
01	O	1	12	0.00650	0.20238	0.44444
01	O	2	12	0.01577	0.09224	1.00000
01	O	3	12	-	-	-
01	A	1	12	0.00642	0.19140	0.00000
01	A	2	12	0.00371	0.04234	0.77778
01	A	3	12	-	-	-
01	M	1	12	0.00590	0.18631	0.11111
01	M	2	11	-	-	-
01	M	3	12	0.00556	0.17253	0.00000
02	O	1	12	0.00537	0.17576	1.00000
02	O	2	12	0.01101	0.01501	0.91111
02	O	3	12	0.01051	0.01451	1.00000
02	A	1	12	0.00521	0.16909	0.00000
02	A	2	12	0.00787	0.16609	0.11111
02	A	3	12	0.00794	0.01350	1.00000
02	M	1	12	0.00571	0.17803	0.00000
02	M	2	12	0.00699	0.18604	0.00000
02	M	3	12	0.00561	0.18310	0.00000
03	O	1	12	0.00907	0.14145	0.11111
03	O	2	12	0.02607	0.01273	0.91111
03	O	3	12	0.00399	0.01158	1.00000
03	A	1	12	0.00514	0.16287	0.00000
03	A	2	12	0.00431	0.15325	0.00000
03	A	3	12	0.00361	0.15504	0.00000
03	M	1	12	0.00363	0.13845	0.60000
03	M	2	12	0.00383	0.15885	0.00000
03	M	3	12	0.00432	0.15848	0.00000
04	O	1	12	0.00496	0.04810	0.80000

World	Arm	Repeat	Batches	Other-nine MAE	Dilute MAE	Dilute coverage
04	O	2	12	0.00697	0.14143	0.66667
04	O	3	12	0.00756	0.11121	0.80000
04	A	1	12	0.00465	0.14272	0.00000
04	A	2	12	0.00471	0.14012	0.00000
04	A	3	12	-	-	-
04	M	1	12	0.00498	0.14234	0.00000
04	M	2	12	0.00769	0.14730	0.00000
04	M	3	12	0.00364	0.10860	0.33333
05	O	1	12	0.00377	0.14630	0.71111
05	O	2	12	0.00818	0.08730	1.00000
05	O	3	12	0.00515	0.13473	0.00000
05	A	1	12	0.00424	0.12640	0.64444
05	A	2	12	0.00383	0.13206	0.00000
05	A	3	12	0.00546	0.12764	0.00000
05	M	1	12	0.00821	0.06752	0.88889
05	M	2	12	0.00502	0.11362	0.00000
05	M	3	12	0.00281	0.10662	0.24444

Complete crystallization process outcomes

All rows have twelve source batches and valid forecasts. Success applies the original joint recommendation criteria to the noiseless retest. All M sessions, and all W02/W04 sessions, belong to the extension cohort. Other O/A sessions belong to the first cohort.

World	Arm	Repeat	Success	Purity MAE	Recovery MAE	Fines MAE
01	O	1	No	0.01514	0.07036	0.49144
01	O	2	Yes	0.01494	0.19041	0.37607
01	O	3	Yes	0.01529	0.05894	0.26897
01	A	1	No	0.06045	0.20875	0.39186
01	A	2	No	0.07045	0.08237	0.35256
01	A	3	Yes	0.00811	0.16938	0.41231
01	M	1	Yes	0.04269	0.07619	0.29191
01	M	2	Yes	0.06712	0.14437	0.17320
01	M	3	Yes	0.04295	0.05451	0.41392
02	O	1	No	0.02065	0.05299	0.34607
02	O	2	Yes	0.02067	0.21933	0.29524
02	O	3	Yes	0.01292	0.15573	0.41107
02	A	1	No	0.00615	0.12426	0.24719
02	A	2	No	0.07982	0.22558	0.37703
02	A	3	No	0.00551	0.11615	0.36440
02	M	1	Yes	0.07148	0.10312	0.51203
02	M	2	Yes	0.07898	0.06437	0.26370
02	M	3	Yes	0.01983	0.17259	0.38024
03	O	1	No	0.00939	0.17261	0.39280
03	O	2	Yes	0.02235	0.05261	0.30280
03	O	3	Yes	0.03235	0.12178	0.33864
03	A	1	No	0.02652	0.15761	0.24197
03	A	2	Yes	0.08319	0.07243	0.35403
03	A	3	Yes	0.03569	0.05836	0.32115
03	M	1	No	0.04944	0.06940	0.30115
03	M	2	Yes	0.04319	0.06980	0.27032
03	M	3	Yes	0.06485	0.05317	0.32115
04	O	1	Yes	0.01027	0.04411	0.24413
04	O	2	No	0.01105	0.11431	0.24150
04	O	3	Yes	0.02029	0.15714	0.26580
04	A	1	Yes	0.09446	0.07348	0.21295
04	A	2	Yes	0.04029	0.10416	0.22004
04	A	3	No	0.00994	0.05034	0.29795

World	Arm	Repeat	Success	Purity MAE	Recovery MAE	Fines MAE
04	M	1	Yes	0.11529	0.04371	0.26628
04	M	2	Yes	0.12113	0.04513	0.19711
04	M	3	Yes	0.08529	0.07505	0.26128
05	O	1	No	0.00534	0.09232	0.14739
05	O	2	No	0.00543	0.24727	0.31748
05	O	3	Yes	0.01443	0.05151	0.32656
05	A	1	Yes	0.01246	0.07328	0.28195
05	A	2	Yes	0.02060	0.08932	0.29656
05	A	3	No	0.06445	0.32577	0.25906
05	M	1	Yes	0.02927	0.11660	0.28520
05	M	2	Yes	0.04528	0.10225	0.32187
05	M	3	Yes	0.01778	0.13283	0.24990

Incomplete chains

All six incomplete chains are retained. Missing stages do not imply a forecast of zero. K1 is the scientific account, Q the prediction, K2 the retrospective, and EQS the structured equilibrium supplement. Five chains lose one or more assessment payloads. The extension W01/M repeat 2 terminates after eleven batches on a resource-accounting validation failure; none of its assessment stages is available.

World	Arm	Repeat	Cohort	Missing stages
05	O	2	First	K2, EQS
01	O	3	First	Q, K2, EQS
01	A	3	First	K1, Q, K2, EQS
04	A	3	First	Q, K2, EQS
03	M	1	Extension	EQS
01	M	2	Extension	K1, Q, K2, EQS

Scope of process cases

Main Figures 2-4 and Extended Data Figures 1-2 connect original public records to executed actions. The equilibrium mapping case is world 4, M, repeat 1; the seed-rescue case world 1, M, repeat 3; the volume-window case world 1, M, repeat 2. The purity comparison retains all three world-4/M sessions. The incorrect cumulative-seed account is world 4, A, repeat 3; the rejection-recognizing counterexample is world 2, A, repeat 2. Risk-choice examples are world 2, A, repeat 3 and world 4, A, repeat 2. These labels locate evidence; they do not define participant subpopulations or prevalence denominators.

All matched quench dispositions

Donor numbers below identify the eighteen first-cohort recommendations in a fixed sorted order. Five worlds are reused for each donor. A dash is a missing final paired effect, with the failed arm retained. Effects are present minus absent and use noiseless fines. They describe the complete pairs, not all planned recipes. Invalid arms are not repaired.

Donor	Source world	Information	Repeat
1	01	Aligned	01
2	01	Opaque	01
3	03	Aligned	01
4	03	Opaque	01
5	05	Aligned	01
6	05	Opaque	01
7	01	Aligned	02
8	01	Opaque	02
9	03	Aligned	02
10	03	Opaque	02
11	05	Aligned	02
12	05	Opaque	02
13	01	Aligned	03

Donor	Source world	Information	Repeat
14	01	Opaque	03
15	03	Aligned	03
16	03	Opaque	03
17	05	Aligned	03
18	05	Opaque	03

Donor	World 1	World 2	World 3	World 4	World 5
1	Absent fails	Absent fails	Absent fails	Absent fails	Absent fails
2	0.00000	0.00000	0.00000	0.00000	0.00000
3	-0.07193	0.08121	-0.06641	-0.03122	-0.11081
4	0.30483	0.25054	0.26820	0.29006	0.00016
5	0.36050	0.75196	0.74491	0.70227	0.74283
6	0.00000	0.00000	0.00000	0.00000	0.00000
7	0.00000	-0.00147	0.00000	0.00000	0.00000
8	0.73643	0.67614	0.65346	0.61596	0.62552
9	Present fails	Present fails	Present fails	Present fails	Present fails
10	Present fails	Present fails	Present fails	Present fails	Present fails
11	0.17751	0.52630	0.52558	0.49988	0.57637
12	0.21897	0.14705	0.14699	0.07543	0.06037
13	0.00558	-0.05076	-0.09236	-0.08477	-0.21882
14	0.71771	0.64406	0.68547	0.62873	0.64872
15	0.60226	0.00022	0.61733	0.55273	0.34794
16	0.54132	0.42409	0.56943	0.54616	0.59024
17	0.57332	0.37074	0.54879	0.55370	0.49501
18	0.43274	0.47053	0.50706	0.52815	0.56957

Retained broad-study tables

The following numerical tables retain the earlier cross-system and targeted-model results without pooling them with the process cohort. Their original descriptive groupings, task-specific response scales and sample units remain unchanged. E, P and S as information loci denote entity, parameter and structure information; P as a system denotes purification. EC is electrochemistry, PA partitioning, RX reaction, EQ equilibrium and C crystallization. Tables with example trajectories refer to the earlier study, not to the new process cases. SD is descriptive variation, not a confidence interval.

Retained table 1: A.1 Complete study matrix

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / E: medium identity	Characterize entity-conditioned responses	12	15	180 / 180
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	**Six system families**		**240**	**3,597 / 3,600**

Table A1. Cross-system study scope (GPT-5.6 Sol, medium). Each campaign has one agent realization. One C campaign exhausts its solvent stock and discards its twelfth vessel, completing eleven final assays. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have scientific account (K1), sealed prediction (Q), and retrospective review (K2) stages; 238 meet the full source-and-assessment specification.

Retained table 2: A.2 Prediction targets and system-specific questions

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / E, P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table A2. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

Tables 3-18 report MAE / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions remain included. PA intervals are nominally 90%; all others are 80%. Response scales and reference targets differ across systems.

Retained table 3: EC, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

Retained table 4: EC, E prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

Retained table 5: PA, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

Retained table 6: EC, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

Retained table 7: EC, E prior: optimization, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

Retained table 8: PA, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

Retained table 9: RX, P prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

Retained table 10: RX, P prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

Retained table 11: RX, S prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

Retained table 12: RX, S prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

Retained table 13: EQ, P prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

Retained table 14: EQ, S prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

Retained table 15: EQ, E prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0128 / 83.7	0.0107 / 85.0	0.0134 / 79.0
Normalized pH	0.0233 / 80.7	0.0278 / 79.0	0.0202 / 82.0
Precipitation signal	0.0084 / 89.3	0.0116 / 92.0	0.0103 / 88.3

Retained table 16: C, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

Retained table 17: C, E prior: delivery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

Retained table 18: P, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

Retained table 19: C.1 Equilibrium: a reversal across public concentration regimes

Query group / arm	Macro MAE	Coverage (%)	Interval width	Source-mean MAE
Other nine / Opaque	0.01008	90.7	0.04706	0.00553
Other nine / Aligned	0.00465	91.3	0.03340	0.00518
Other nine / MisIndexed	0.00439	89.6	0.02798	0.00494
Three most dilute / Opaque	0.01833	93.3	0.15676	0.16961
Three most dilute / Aligned	0.15862	2.2	0.08084	0.16833
Three most dilute / MisIndexed	0.14397	13.8	0.09633	0.16836

Table C1. EQ/P subgroup results. Each row averages five worlds; rows reuse those worlds. The source-mean predictor is an additional post hoc diagnostic and obtains no new experiments. Its poor dilute-query performance shows why retaining the plateau is insufficient; its usefulness on the other nine depends on their response distribution.

Retained table 20: C.2 Crystallization: joint response changes under the same treatment

Budget / arm	Recovery MAE	Purity MAE	Recovery improves and purity worsens vs Opaque
12 / Opaque	0.16322	0.00952	Reference
12 / Aligned	0.06055	0.04969	5 / 5 worlds
12 / MisIndexed	0.08923	0.09166	5 / 5 worlds
24 / Opaque	0.09107	0.01975	Reference
24 / Aligned	0.06839	0.03983	2 / 5 worlds
24 / MisIndexed	0.07863	0.05080	3 / 5 worlds

Table C2. Joint directional comparisons use both response errors from the same world, arm and budget. At twelve batches, each joint direction also holds in all four pairs whose two campaigns conform to the source-assay contract. The direction weakens at 24 batches. A treatment can therefore be helpful on one response and harmful on another without implying a budget-invariant tradeoff.

Retained table 21: C.4 Original equilibrium campaigns: predictions and research chronology

World / arm	Public Q explanation	Dissociation [80% interval]	Reference
1 / Opaque	Piecewise weak acid	0.6990 [0.250, 0.900]	0.7160
1 / Aligned	Plateau / weak trend	0.1040 [0.044, 0.225]	0.7160
1 / MisIndexed	Plateau / weak trend	0.1092 [0.0772, 0.1412]	0.7160
2 / Opaque	Piecewise weak acid	0.6290 [0.450, 0.800]	0.6546
2 / Aligned	Plateau / weak trend	0.0804 [0.053, 0.108]	0.6546
2 / MisIndexed	Plateau / weak trend	0.0806 [0.0556, 0.1056]	0.6546
3 / Opaque	Piecewise weak acid	0.5630 [0.430, 0.700]	0.5797
3 / Aligned	Plateau / weak trend	0.1100 [0.045, 0.190]	0.5797
3 / MisIndexed	Mixed weak acid / plateau	0.4600 [0.100, 0.750]	0.5797
4 / Opaque	Qualitative departure	0.5200 [0.150, 0.880]	0.5173
4 / Aligned	Plateau / weak trend	0.0724 [0.040, 0.105]	0.5173
4 / MisIndexed	Plateau / weak trend	0.0962 [0.040, 0.450]	0.5173
5 / Opaque	Piecewise weak acid	0.4130 [0.200, 0.700]	0.4480
5 / Aligned	Plateau / weak trend	0.0790 [0.035, 0.130]	0.4480
5 / MisIndexed	Plateau / weak trend	0.0681 [0.040, 0.096]	0.4480

Table C3. Public extrapolation accounts and original same-context predictions. Categories are a single-author retrospective reading of public explanations, not internal reasoning traces or independent causal labels. Four Opaque accounts use piecewise weak-acid reasoning; the fifth permits a qualitative departure. All five Aligned accounts extend a plateau or weak trend. The third-world MisIndexed mixed account is a counterexample to treating all dossier conditions alike.

Retained table 22: E.3 Complete crystallization baseline and interval comparisons

Table E1. Crystallization prediction errors and baseline comparisons.

Response	Agent MAE	Mean MAE	Nearest MAE	Wins: mean	Wins: nearest
Recovery	0.09185	0.17135	0.17308	26/30	26/30
Purity	0.04354	0.00482	0.00905	0/30	4/30
Size index	0.06923	0.02345	0.02708	4/30	7/30
Fines fraction	0.29000	0.31651	0.32426	21/30	20/30

MAE is averaged equally over all thirty campaigns in each response's original scale; errors are not ranked across responses. Recovery, purity and fines are fractions, while size is the declared size index. Mean and Nearest denote the public observation-mean and nearest-neighbour predictors. Wins count strictly lower agent MAE; neither comparison has ties. The source-assay shortfall remains included, and the five world instances are reused across arms and independent budget sessions.

Retained table 23: E.3 Complete crystallization baseline and interval comparisons

Table E2. Purity interval coverage and width by information arm and budget.

Budget	Information arm	Campaigns	Coverage (%)	Mean width (pp)
12	Opaque	5	98.3	6.81
12	Aligned	5	60.0	12.94
12	MisIndexed	5	23.3	15.53
24	Opaque	5	95.0	8.59
24	Aligned	5	73.3	12.37
24	MisIndexed	5	55.0	12.31

The original intervals have nominal 80% coverage and concern the pre-final-assay response target. Coverage and interval width are averaged over each campaign's twelve queries and then over its five-world arm-budget group. Each row contains sixty query targets but only five world clusters; queries are not independent world replicates. Width is expressed in percentage points of purity. All original intervals and source failures remain unchanged.

Retained table 24: E.4 Complete research-objective comparison

Table E3. All joint goal outcomes.

Higher retest	Lower score MAE	EC pairs	RX pairs
Optimization	Optimization	13	4
Optimization	Discovery	13	5
Discovery	Optimization	1	4
Discovery	Discovery	3	17

Both endpoints favour the same campaign in 16/30 EC and 21/30 RX pairs, and opposite campaigns in 14/30 and 9/30. Strict signs define these descriptive counts; near-zero differences are not thereby significant. Each system reuses five worlds. RX score MAE favours optimization in 8/30 pairs; its separate six-response macro MAE does so in 6/30.

Retained table 25: E.5 Strata and recovery sensitivity

Table E4. Endpoint orderings within design strata.

System	Factor	Level	Pairs	Same	Opposite
EC	World	W01	6	3	3
EC	World	W02	6	4	2
EC	World	W03	6	3	3
EC	World	W04	6	5	1
EC	World	W05	6	1	5
EC	Budget	12	15	7	8
EC	Budget	24	15	9	6
EC	Arm	Aligned	10	6	4
EC	Arm	MisIndexed	10	5	5
EC	Arm	Opaque	10	5	5
EC	Locus	E	30	16	14
RX	World	W01	6	4	2
RX	World	W02	6	4	2
RX	World	W03	6	3	3
RX	World	W04	6	4	2
RX	World	W05	6	6	0
RX	Budget	12	30	21	9
RX	Arm	Aligned	10	6	4
RX	Arm	MisIndexed	10	9	1
RX	Arm	Opaque	10	6	4
RX	Locus	P	15	12	3

System	Factor	Level	Pairs	Same	Opposite
RX	Locus	S	15	9	6

Same means that the higher-retest campaign also has lower MAE; opposite means it has higher MAE. These overlapping strata do not add independent tests. Budget rows compare goals at a fixed budget, not a twelve-to-twenty-four-batch contrast. E denotes entity priors, P parameter priors and S structural priors.

Retained table 26: E.6 All batches of the selected electrochemical pair

Table E5. Discovery source batches.

Batch	Pair	V	mA	Electrolysis / s	Score
1	S0/E0	-1.5	100	1200	0.000000
2	S1/E0	-1.5	100	1200	0.167530
3	S2/E0	-1.5	100	1200	0.000000
4	S3/E0	-1.5	100	1200	0.050903
5	S0/E0	0.5	100	1200	0.509008
6	S0/E0	1.5	100	1200	0.044293
7	S0/E0	2.5	100	1200	0.030644
8	S0/E1	0.5	100	1200	0.000000
9	S0/E2	0.5	100	1200	0.482517
10	S0/E3	0.5	100	1200	0.360679
11	S0/E2	0.5	100	1200+13200	0.606942
12	S0/E2	0.5	500	1200+13200	0.710658

Retained table 27: E.6 All batches of the selected electrochemical pair

Table E6. Optimization source batches.

Batch	Pair	V	mA	Electrolysis / s	Score
1	S0/E0	1	100	3600	0.535704
2	S1/E1	-1.5	300	7200	0.282124
3	S2/E2	1.5	300	7200	0.674932
4	S3/E3	1.5	300	7200	0.313717
5	S2/E0	1.5	300	7200	0.033808
6	S0/E2	1.5	300	7200	0.042443
7	S2/E2	1	300	7200	0.570485
8	S2/E2	2	300	7200	0.000000
9	S2/E2	1.5	500	7200	0.722737
10	S2/E2	1.5	500	10000	0.731956
11	S2/E2	1.4	500	9000	0.714786
12	S2/E2	1.5	400	12000	0.749194

Both campaigns select batch 12. Independent retests are 0.717587 for discovery and 0.756122 for optimization; twelve-query score MAEs are 0.088394 and 0.229397.

Retained table 28: E.8 Complete optimization forecasts

Table E7. Query settings, sealed forecasts and reference scores.

Q	Pair	V	mA	s	Forecast	80% interval	Reference
01	S0/E0	-0.8	100	600	0.050	0.000-0.180	0.032659
02	S0/E0	0.8	100	600	0.230	0.040-0.480	0.349491
03	S0/E0	-0.8	100	7200	0.250	0.080-0.440	0.358985
04	S0/E0	0.8	100	7200	0.570	0.430-0.680	0.572012

Q	Pair	V	mA	s	Forecast	80% interval	Reference
05	S0/E1	-0.8	100	7200	0.080	0.000-0.250	0.362359
06	S0/E1	0.8	100	7200	0.100	0.000-0.290	0.627599
07	S0/E3	-0.8	100	7200	0.070	0.000-0.230	0.347853
08	S0/E3	0.8	100	7200	0.090	0.000-0.280	0.542955
09	S2/E0	-0.8	100	7200	0.060	0.000-0.210	0.288226
10	S2/E0	0.8	100	7200	0.070	0.000-0.240	0.494376
11	S0/E0	-0.8	500	7200	0.170	0.040-0.360	0.276451
12	S0/E0	0.8	500	7200	0.550	0.390-0.680	0.344881

Score MAE is 0.229397; interval coverage is 5/12. The accurate Q04 prediction is retained alongside the large underestimates at Q06, Q08 and Q10. Source-to-query changes involve amount, volume and electrical settings, so these contrasts do not isolate one changed variable. Twelve query outcomes do not establish population calibration. No intermediate sealed prediction checkpoints were collected, so final prediction errors cannot be plotted as a within-session learning curve.

Retained table 29: F.1 Original intervals and campaign-specific observations

Table F1. Dissociation predictions at the most dilute recipe, with all values in percent.

World / arm	Source range	Point prediction	80% interval	Reference mean
1 / Opaque	6.18-8.34	69.90	[25.00, 90.00]	71.60
1 / Aligned	6.20-9.23	10.40	[4.40, 22.50]	71.60
1 / MisIndexed	6.11-9.25	10.92	[7.72, 14.12]	71.60
2 / Opaque	6.17-8.57	62.90	[45.00, 80.00]	65.46
2 / Aligned	6.12-8.06	8.04	[5.30, 10.80]	65.46
2 / MisIndexed	6.23-7.98	8.06	[5.56, 10.56]	65.46
3 / Opaque	5.94-8.22	56.30	[43.00, 70.00]	57.97
3 / Aligned	6.34-8.02	11.00	[4.50, 19.00]	57.97
3 / MisIndexed	6.16-7.94	46.00	[10.00, 75.00]	57.97
4 / Opaque	6.19-7.91	52.00	[15.00, 88.00]	51.73
4 / Aligned	5.82-8.14	7.24	[4.00, 10.50]	51.73
4 / MisIndexed	5.81-8.22	9.62	[4.00, 45.00]	51.73
5 / Opaque	5.49-7.48	41.30	[20.00, 70.00]	44.80
5 / Aligned	6.44-7.99	7.90	[3.50, 13.00]	44.80
5 / MisIndexed	6.58-7.79	6.81	[4.00, 9.60]	44.80

The recipe contains 1 micromole in 75 mL (nominal concentration 13.3 μM). The displayed source ranges are specific to each campaign. World 3 MisIndexed retains its mixed-explanation departure from the plateau. Public explanation categories appear in retained table 21 above.

Retained table 30: F.2 Sol interval and regime comparisons

Supplementary Table F2. GPT-5.6 Sol equilibrium predictions across test regions.

Arm	Remaining nine: MAE	Lowest three: MAE	Remaining nine: coverage	Lowest three: coverage
Opaque	0.01008	0.01833	90.7%	93.3%
Aligned	0.00465	0.15862	91.3%	2.2%
MisIndexed	0.00439	0.14397	89.6%	13.8%

Values average five worlds and three response targets per group: normalized pH, dissociation fraction and precipitation signal. Coverage uses original 80% prediction intervals, with 675 reference-containment judgments per arm for the remaining nine conditions and 225 for the three lowest-concentration conditions; these judgments are not independent world replicates. The concentration grouping is post hoc and exploratory; the remaining nine conditions also include boundaries.

Retained table 31: F.3 Complete five-model comparison

Supplementary Table F3. Equilibrium predictions from five medium model configurations.

Model	Arm	Remaining nine: MAE	Lowest three: MAE	Lowest three: coverage
GPT-5.5	Opaque	0.00647	0.16325	6.7%
GPT-5.5	Aligned	0.00638	0.16146	2.7%
GPT-5.5	MisIndexed	0.00461	0.13604	13.3%
GPT-5.6 Luna	Opaque	0.09867	0.15240	48.4%
GPT-5.6 Luna	Aligned	0.04997	0.13319	31.1%
GPT-5.6 Luna	MisIndexed	0.05494	0.12556	44.4%
GPT-5.6 Terra	Opaque	0.03268	0.14611	67.1%
GPT-5.6 Terra	Aligned	0.01545	0.11466	26.2%
GPT-5.6 Terra	MisIndexed	0.01369	0.12130	41.8%
GPT-5.6 Sol	Opaque	0.01008	0.01833	93.3%
GPT-5.6 Sol	Aligned	0.00465	0.15862	2.2%
GPT-5.6 Sol	MisIndexed	0.00439	0.14397	13.8%
GPT-6 Astra	Opaque	0.00244	0.04040	82.7%
GPT-6 Astra	Aligned	0.00384	0.03853	82.7%
GPT-6 Astra	MisIndexed	0.00234	0.01857	92.4%

Each row contains five independent source campaigns, equally weighting the five worlds; worlds recur across information conditions and models. Each campaign predicts pH/14, dissociation fraction and precipitation signal for twelve queries. The three lowest-concentration conditions are the same fixed query group; the remaining nine also include boundary and unexplored conditions. Coverage concerns original 80% intervals and five reference observations, not confidence intervals or independent model replications. All three arms are retained; lower MisIndexed error does not itself establish a causal benefit of incorrect knowledge.

Retained table 32: G.1 Equilibrium reversal by response

Table G1. Original prediction MAE by response and concentration group.

Response	Group	Opaque	Aligned	MisIndexed	Lower / higher
pH / 14	Other nine	0.00261	0.00269	0.00254	2 / 3 of 5
pH / 14	Dilute three	0.00448	0.05576	0.05230	0 / 5 of 5
Dissociation fraction	Other nine	0.00839	0.00709	0.00660	3 / 2 of 5
Dissociation fraction	Dilute three	0.02640	0.28229	0.23861	0 / 5 of 5
Precipitation signal	Other nine	0.01923	0.00417	0.00403	5 / 0 of 5
Precipitation signal	Dilute three	0.02411	0.13783	0.14099	0 / 5 of 5

The final column counts worlds with lower/higher Aligned MAE than Opaque. Signed response differences divided by three sum to the macro-error difference. They quantify the contribution on the declared normalized scales; they do not establish a shared failure mechanism or physical comparability of targets.

Retained table 33: G.2 Separating purity offset from variation error

Table G2. Purity offset and variation diagnostics.

For each campaign, prediction error decomposes as MSE $= \overline{p-y}^{\,2} + \frac{1}{12}\sum_i[(p_i-\bar{p})-(y_i-\bar{y})]^2$. Centred RMSE is the square root of the second term. Demeaning uses withheld references for diagnosis, not as an available predictor.

Batches	Arm	N	Bias	cRMSE	Ref. SD	Pred. SD	Offset share
Both	All	30	-4.289	1.889	0.598	1.968	87.6%
12	Opaque	5	-0.806	0.970	0.598	1.031	45.3%
12	Aligned	5	-4.920	1.837	0.598	1.791	89.8%
12	MisIndexed	5	-9.166	2.843	0.598	2.977	91.3%
24	Opaque	5	-1.816	1.682	0.598	1.649	57.4%

Batches	Arm	N	Bias	cRMSE	Ref. SD	Pred. SD	Offset share
24	Aligned	5	-3.948	1.797	0.598	1.983	85.6%
24	MisIndexed	5	-5.080	2.206	0.598	2.376	85.3%

N is the campaign count; cRMSE denotes centred RMSE. Bias, cRMSE and both standard deviations are campaign means in percentage points. Offset share is the ratio of summed squared campaign biases to summed campaign MSEs, with equal query counts; it is not the mean campaign fraction. The median campaign fraction is 77.6%. Mean bias is negative in 30 of 30 campaigns. Centred RMSE exceeds reference SD in 29 of 30. These comparisons describe forecast outputs and do not identify an internal revision process. No new agent or simulator calls were made.

Retained table 34: H.2 A concrete three-arm information example

Table H1. Public prior content in one matched physical world.

Supplied field	Opaque	Aligned	MisIndexed
Effective pKa interval (80%)	Absent	[4.6094, 4.7094]	[5.3294, 5.4294]
Dilution change in pH/14	Absent	+0.003494987	+0.000875889
Dilution change in dissociation	Absent	+0.008187416	+0.001884676
Nominal confidence	Absent	0.8	0.8

The common dilution anchor is 0.001 mol acid at 298.15 K, with volume increasing from 0.018 to 0.054 L. Both supplied records identify the source as an "independent bounded archival fit" and qualify the claim as a "local effective relationship; not a universal aqueous-chemistry law". The pKa intervals above are rounded for display; the input contains their full precision. Aligned therefore supplies correct partial local information, not the evaluator's complete process equations, query answers or an instruction to enter the most dilute test region. The experiment contrasts the resulting complete research processes, including possible changes to evidence acquisition and interpretation.

Executed process relationships

The following equations describe the retained virtual-world laws, not mechanisms inferred by the participants. Historical final-assay and current retest recovery normalizations are distinct; current process outcomes follow the Methods above.

Equilibrium response model

The parameter-prior study uses a bounded monoprotic acid-base model followed by a precipitation calculation. These are the process relations used for the retained campaigns and references. They are distinct from explanations proposed by the agents. In particular, an agent's proposed active-pool cap is not the implemented source of the dissociation plateau.

Let n be the current ledger amount of the initial reactant, V the current sample volume and $C = n/V$. The background-ion amounts are $n_+ = \min(0.020, 0.15n)$ and $n_- = \min(0.020, 0.08n)$, in moles. With $H = [\mathrm{H^+}]$, $K_a = 10^{-\mathrm{p}K_a}$ and activity-ratio factor g, the solver obtains

$$[\mathrm{A^-}] = \frac{CK_a}{K_a + gH}, \qquad [\mathrm{OH^-}] = \frac{K_w(T)}{H},$$

subject to charge balance,

$$H + \frac{n_+}{V} - \frac{K_w(T)}{H} - [\mathrm{A^-}] - \frac{n_-}{V} = 0.$$

The reported responses are $\mathrm{pH}/14 = -\log_{10}(H)/14$ and dissociation fraction $\alpha = [\mathrm{A^-}]/C$, clipped to the public unit interval. In the uncapped background-ion range, charge balance implies $\alpha = 0.07 + H/C - K_w/(HC)$. At sufficiently high concentration the last two terms are small, giving a local plateau near 7%; dilution makes their contribution appreciable. The plateau and departure therefore arise under the same fixed equations. The five effective pKa values span 4.6594 to 5.3794 in increments of 0.18; the default activity-ratio factor is one.

Precipitation is calculated sequentially from effective ion inventories $a = n_+ + 0.10n_{\mathrm{A^-}}$ and $b = n_- + 0.08n_{\mathrm{HA}}$. When supersaturated, a 1:1 removal x, bounded by the limiting inventory, satisfies $(a - x)(b - x)/V^2 = K_{sp}$, with default

$K_{sp} = 1.8 \times 10^{-10}$. The public signal is x/n, clipped to $[0, 1]$. This hook does not feed precipitation back into the acid-base solution. Thus the three public responses are related but should not be interpreted as three independent measurements of one failure mechanism.

Nominal concentration in the retained source comparisons is a recipe descriptor: total nominal reagent additions divided by total solvent additions. The solver instead uses the ledger state at measurement, after any sample consumption and other operations. The scatterplot combines distinct recipes and histories; it is not a controlled one-dimensional titration curve. The three most dilute test recipes have nominal concentrations of approximately 1.33×10^{-4}, 1.33×10^{-5} and 1.67×10^{-4} M. No source assay in these fifteen Sol campaigns reaches those concentrations.

Crystallization response model

The retained crystallization block composes upstream reaction, thermal operations, seeding, crystal nucleation and growth, filtration and measurement. It uses the latent-material process family and its linear supersaturation-dependent impurity transfer. Solid product, solid impurity and the particle population persist across operations; dissolved and solid inventories are accounted for separately.

The temperature-dependent solubility follows

$$c^*(T) = c^*_{\text{ref}} \exp\left[-\frac{\Delta H}{R}\left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right],$$

with $T_{\text{ref}} = 298.15$ K, $\Delta H = 20{,}000$ J mol^{-1} and a declared temperature domain of 250-430 K. World and solvent properties scale the reference solubility and kinetics. For relative supersaturation $s = \max(C/c^* - 1, 0)$, primary nucleation scales as $B = k_b s^2$ and linear growth as $G = k_g s$. Before world and solvent multipliers, the coefficients are 2×10^7 L^{-1} s^{-1} for nucleation and 2×10^{-8} m s^{-1} for growth. Nucleation and growth consume available dissolved target and update the population distribution; they are not independent score adjustments.

Each crystallized target increment transfers impurity according to

$$\Delta n_{I,s} = \min(n_{I,l},\ \Delta n_{P,s}\, k_{\text{occ}}(1 + 0.5s)),$$

where $k_{\text{occ}} = 0.02$ times the solvent's occlusion multiplier. The four nominal solvent multipliers are 1.20, 0.80, 0.65 and 1.05; world-fixed material residuals further modulate these properties. Filtration retains 96% of target solids and 92% of impurity solids. The final-assay purity is the solid-target mole fraction $n_{P,s}/(n_{P,s} + n_{I,s})$. Seed contributes to target purity but is subtracted in the full-process recovery numerator; final-assay recovery is normalized by the initial reagent charge.

These relations allow thermal and seeding histories to alter the amount and size distribution of crystals while impurity incorporation remains modest. They explain how recovery and particle responses can vary without requiring a comparable purity change. The approximately 98.5% purity level is an empirical property of the retained source and query conditions, not a hard-coded value or a universal invariant of the model. The present analysis does not isolate individual process contributions to that stability. The retained diagnostics test whether the forecasts preserve the level and variation of the retained purity references.